

```
In [3]: import numpy as np
import pandas as pd
```

DataSet Information

1. The topic of our prject is biology, sociology, and psychology. Our title is: Human Interaction
2. We obtained the data set from Kaggle: (insert link)
3. This data set studies how different factors contribute to an individuals personality.

Brief introduction summarizing dataset.

```
In [4]: human_interaction = pd.read_csv("personality_dataset.csv")
human_interaction
```

```
Out[4]:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 8 columns

Basic Data Exploration

```
In [5]: print(f'The first 5 rows of the data set:')
human_interaction.head(5)
```

```
Out[5]: The first 5 rows of the data set:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained_
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No

```
In [6]: print(f'The last 5 rows of the data set:')
        human_interaction.tail(5)
```

Out[6]: The last 5 rows of the data set:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

```
In [7]: print(f'There are {human_interaction.shape} rows and columns:')
```

Out[7]: There are (2900, 8) rows and columns:

```
In [8]: print(f'The column names are:')
        human_interaction.columns
```

Out[8]: The column names are:

```
Index(['Time_spent_Alone', 'Stage_fear', 'Social_event_attendance',
       'Going_outside', 'Drained_after_socializing', 'Friends_circle_size',
       'Post_frequency', 'Personality'],
      dtype='object')
```

Data Type and Structure

```
In [9]: print(f'The data types are:')
        human_interaction.info()
```

Out[9]: The data types are:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2900 entries, 0 to 2899
Data columns (total 8 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   Time_spent_Alone                     2900 non-null   float64
 1   Stage_fear                           2900 non-null   object
 2   Social_event_attendance              2900 non-null   float64
 3   Going_outside                        2900 non-null   float64
 4   Drained_after_socializing            2900 non-null   object
 5   Friends_circle_size                  2900 non-null   float64
 6   Post_frequency                       2900 non-null   float64
 7   Personality                           2900 non-null   object
dtypes: float64(5), object(3)
memory usage: 181.4+ KB
```

```
In [10]: human_interaction.select_dtypes(include='number')
```

Out[10]:

	Time_spent_Alone	Social_event_attendance	Going_outside	Friends_circle_si
0	4.0	4.0	6.0	13.0
1	9.0	0.0	0.0	0.0
2	9.0	1.0	2.0	5.0
3	0.0	6.0	7.0	14.0

	Time_spent_Alone	Social_event_attendance	Going_outside	Friends_circle_si
4	3.0	9.0	4.0	8.0
...	...	...	...	...
2895	3.0	7.0	6.0	6.0
2896	3.0	8.0	3.0	14.0
2897	4.0	1.0	1.0	4.0
2898	11.0	1.0	3.0	2.0
2899	3.0	6.0	6.0	6.0

2900 rows x 5 columns

Some of the coluns, such as tage_fear and Drained_after_socializing columns should be converted to numerical to test for a correlational analysis so that we may study the personalit type of the individual.

Descriptive Summary

```
In [11]: hi_summary=human_interaction.describe()
print(hi_summary)
hi_mean=human_interaction['Time_spent_Alone'].mean()
print(hi_mean)
hi_median=human_interaction['Time_spent_Alone'].median()
print(hi_median)
hi_range=human_interaction['Time_spent_Alone'].max()+human_interaction['T
ime_spent_Alone'].min()
print(hi_range)
hi_std=human_interaction['Time_spent_Alone'].std()
print(hi_std)
```

```
Out[11]:
```

	Time_spent_Alone	Social_event_attendance	Going_outside	\
count	2900.000000	2900.000000	2900.000000	
mean	4.505816	3.963354	3.000000	
std	3.441180	2.872608	2.221597	
min	0.000000	0.000000	0.000000	
25%	2.000000	2.000000	1.000000	
50%	4.000000	3.963354	3.000000	
75%	7.000000	6.000000	5.000000	
max	11.000000	10.000000	7.000000	

	Friends_circle_size	Post_frequency
count	2900.000000	2900.000000
mean	6.268863	3.564727
std	4.232340	2.893587
min	0.000000	0.000000
25%	3.000000	1.000000
50%	5.000000	3.000000
75%	10.000000	6.000000
max	15.000000	10.000000

4.50581600281988
4.0
11.0
3.4411804188954207

The size of the mean in 6.27 and the standard deviation is 4.23. This shows there is variability within the data. When looking at the amount of friends within each individual, some have 6 and

others have 0 and others have 15 friends. While the range is wide, it exemplifies the variety that lies within our data. There are absolutely negative values or missing values within the numerical values.

Missing or Duplicate Data

```
In [12]: missing = human_interaction.isnull().sum()
print("Missing values in each column:\n")
print(missing)
```

Out[12]: Missing values in each column:

```
Time_spent_Alone      0
Stage_fear            0
Social_event_attendance 0
Going_outside         0
Drained_after_socializing 0
Friends_circle_size   0
Post_frequency        0
Personality           0
dtype: int64
```

```
In [13]: duplicates = human_interaction.duplicated().sum()
print(f"\nThere are {duplicates} duplicate rows in the dataset.")
```

Out[13]: There are 402 duplicate rows in the dataset.

Data Wrangling

```
In [14]: hi_new=human_interaction.copy()
hi_new
```

Out[14]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows × 8 columns

```
In [15]: def drained(Time_spent_Alone):
if pd.isnull(Time_spent_Alone):
    return 'N/A'
elif Time_spent_Alone < 5:
```

```

        return 1
    else:
        return 2

def fright(Friends_circle_size):
    if pd.isnull(Friends_circle_size):
        return 'N/A'
    elif Friends_circle_size <=5:
        return 2
    else:
        return 1
print

```

Out[15]: <function print(*args, sep=' ', end='\n', file=None, flush=False)>

In [16]:

```

hi_new['Drained_after_events']=hi_new['Time_spent_Alone'].apply(drained)
hi_new['Stage_fright']=hi_new['Friends_circle_size'].apply(fright)
hi_new

```

Out[16]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 10 columns

In [17]:

```

hi_new.head(11)

```

Out[17]:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
5	1.0	No	7.0	5.0	No
6	4.0	No	9.0	3.0	No
7	2.0	No	8.0	4.0	No
8	10.0	Yes	1.0	3.0	Yes
9	0.0	No	8.0	6.0	No
10	3.0	No	9.0	6.0	No

Drained after events, our new column, represents the amount of time a person rates their time being spent alone. 5 hours alone is rated as an individual who is an extrovert or isn't easily drained.

Filtering the Data

```
In [18]: hi_filtered1=hi_new.copy()
hi_filtered1
```

```
Out[18]:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 10 columns

```
In [19]: hi_filtered1.drop(columns=['Drained_after_events'], inplace=True,
errors='ignore')
print('\n\n Data after dropping Drained after events column:\n\n')
hi_filtered1
```

```
Out[19]:
```

Data after dropping Drained after events column:

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drain
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No
...	...	...	...	...	...
2895	3.0	No	7.0	6.0	No
2896	3.0	No	8.0	3.0	No
2897	4.0	Yes	1.0	1.0	Yes
2898	11.0	Yes	1.0	3.0	Yes
2899	3.0	No	6.0	6.0	No

2900 rows x 9 columns

Unique Values And Categories

```
In [20]: unique_values=hi_new['Stage_fear'].unique()
print(f'Are there any unique values in the Stage fear column?')
```

```
{unique_values}')
```

Out[20]: Are there any unique values in the Stage fear column? ['No' 'Yes']

```
In [21]: unique_values=hi_new['Drained_after_socializing'].unique()
print(f'Are there any unique values in the Drained after socializing
column? {unique_values}')
```

Out[21]: Are there any unique values in the Drained after socializing column? ['No' 'Yes']

```
In [22]: unique_values2=hi_new['Personality'].unique()
print(f'Are there any unique values in the Personality column?
{unique_values2}')
```

Out[22]: Are there any unique values in the Personality column? ['Extrovert' 'Introvert']

The categorical variable that seems the most important for grouping is the Personality column while the Drained_after_socialization and Stage_fear columns are better when looking at correlation analysis.

Grouping and Aggregation

```
In [23]: personality_by_stage_flight=hi_new.groupby('Personality').agg({
    'Time_spent_Alone': 'mean',
    'Friends_circle_size': 'mean',
    'Social_event_attendance': 'mean'
})
personality_by_stage_flight
```

Out[23]:

	Time_spent_Alone	Friends_circle_size	Social_event_attendance
Personality			
Extrovert	2.122869	9.095744	5.977850
Introvert	7.027444	3.277465	1.831621

When looking at this data, it is evident that there is a difference in habit when looking at the differing personality types. Introverts spend less time with people while extroverts have three times more of events and hold longer friendships as well.

Summary and Conclusion

An interesting thing about this data set is the contrast between introverts and extroverts and the difference was the differences in social energy and the balance of time as well. There was a significant difference in time spent in friendship groups and the amount of social events that were attended. When looking at the column Friends_circle_size, the data distribution shows that the post_frequency column is high, showing that just because someone may have high social engagement does not mean they have high social circles. Within our data, there is a small amount of neutral or ambivert categories. This creates questions about whether this was meant to happen on

puprose or if individuals taking the test had to pick between the only given options which were Introvert and Extrovert.

```
In [24]: hi_filtered1.to_csv('Personality_Dataset_filtered.csv', index=False)
```

```
In [25]: hi_filtered1.head()
```

```
Out [25]:
```

	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_outside	Drained_
0	4.0	No	4.0	6.0	No
1	9.0	Yes	0.0	0.0	Yes
2	9.0	Yes	1.0	2.0	Yes
3	0.0	No	6.0	7.0	No
4	3.0	No	9.0	4.0	No

```
In [26]: hi_filtered1.to_csv('/home/user/Personality_Dataset_filtered.csv',
index=False)
```

```
In [27]: import os
os.listdir('/home/user')
```

```
Out [27]: ['.before_class_mar16.ipynb.sage-jupyter2',
'.config',
'students.csv',
'.snapshots',
'.local',
'.npm',
'March_23_Project2Data.ipynb',
'.jupyter',
'Projects',
'Practice_Problems_Exam.ipynb',
'.March_23_Project2Data.ipynb.sage-jupyter2',
'.ssh',
'Question_Bank',
'Project2_Part1_InClass.ipynb',
'.bash_profile',
'Personality_Dataset_filtered.csv',
'.Practice_Problems_Exam.ipynb.sage-jupyter2',
'In_class_Practice',
'.smc',
'.ipython',
'.Project2_Part1_InClass.ipynb.sage-jupyter2',
'.cache',
'Project 1 Part II Folder',
'Daily_Challenge',
'.bashrc',
'before_class_mar16.ipynb']
```

```
In [28]: hi_filtered1.to_csv('test.csv', index=False)
```

PROJECT 2 PART 2 - DATA VISUALIZATION

```
In [29]: import numpy as np
import pandas as pd
```



```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [30]: hi_filtered=pd.read_csv("Personality_Dataset_filtered1.csv")
hi_filtered
```

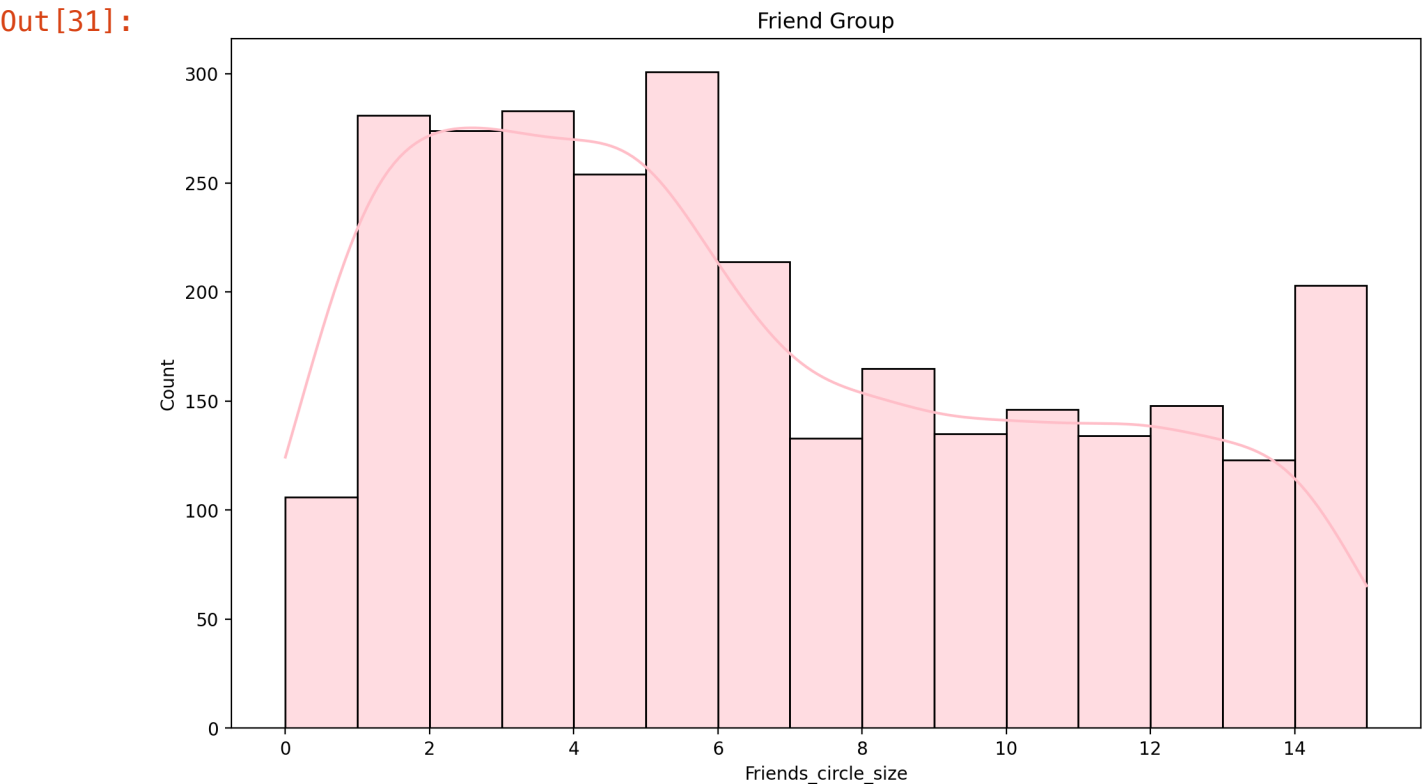
Out [30]:

	Unnamed: 0	Time_spent_Alone	Stage_fear	Social_event_attendance	Going_out
0	0	4.0	No	4.0	6.0
1	1	9.0	Yes	0.0	0.0
2	2	9.0	Yes	1.0	2.0
3	3	0.0	No	6.0	7.0
4	4	3.0	No	9.0	4.0
...	...	...	...	...	...
2895	2895	3.0	No	7.0	6.0
2896	2896	3.0	No	8.0	3.0
2897	2897	4.0	Yes	1.0	1.0
2898	2898	11.0	Yes	1.0	3.0
2899	2899	3.0	No	6.0	6.0

2900 rows x 11 columns

Section 1:

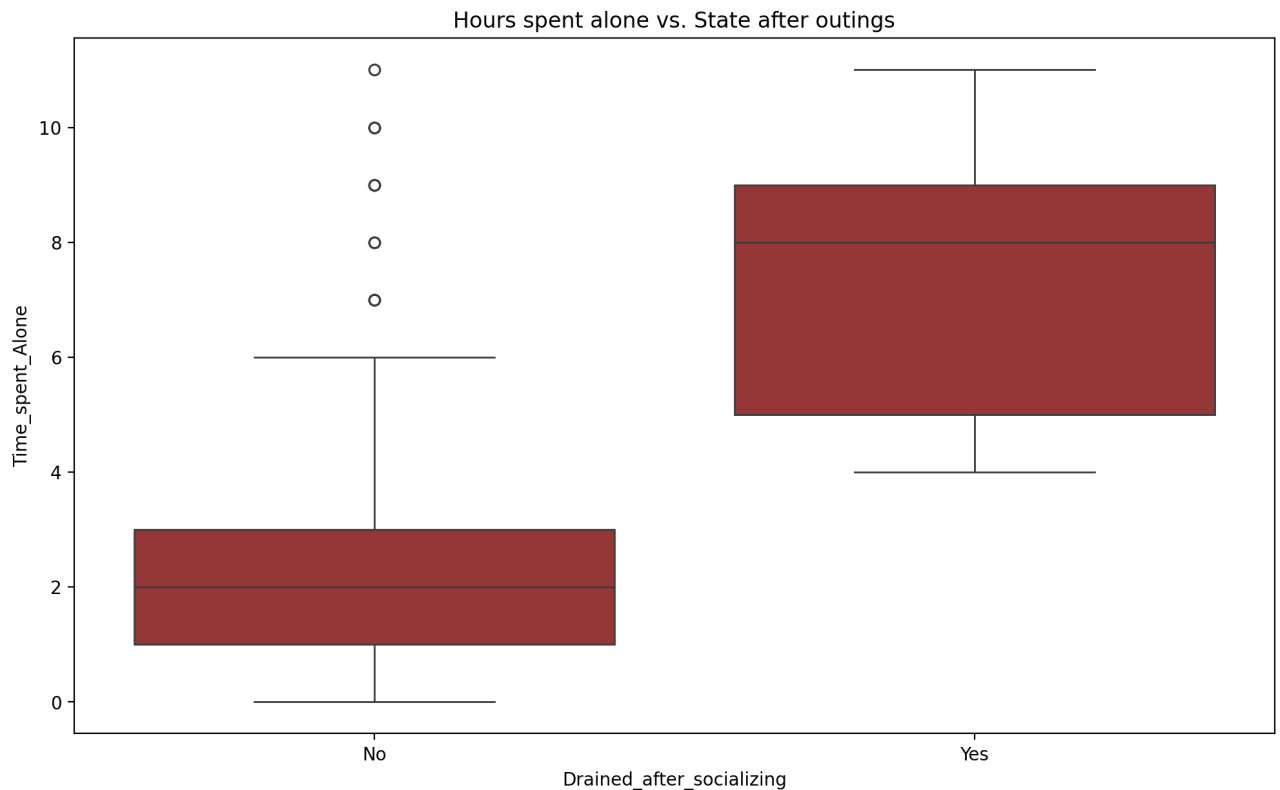
```
In [31]: #histplot
sns.histplot(hi_filtered['Friends_circle_size'], bins=15, color='pink',
kde=True)
plt.title("Friend Group")
plt.savefig('Project2_Graphs/hist_friend_size.jpg')
plt.show()
```



When looking at the histogram, it shows the distribution of values regarding the size of friend groups. The results show that friend group size has a spread that is moderate such as having friend group sizes ranging from 2-6 people.

```
In [32]: sns.boxplot(data=hi_new, x='Drained_after_socializing',
y='Time_spent_Alone', color='brown')
plt.title('Hours spent alone vs. State after outings')
plt.savefig('Project2_Graphs/Alone_time_vs._State_after_outings.jpg')
plt.show()
```

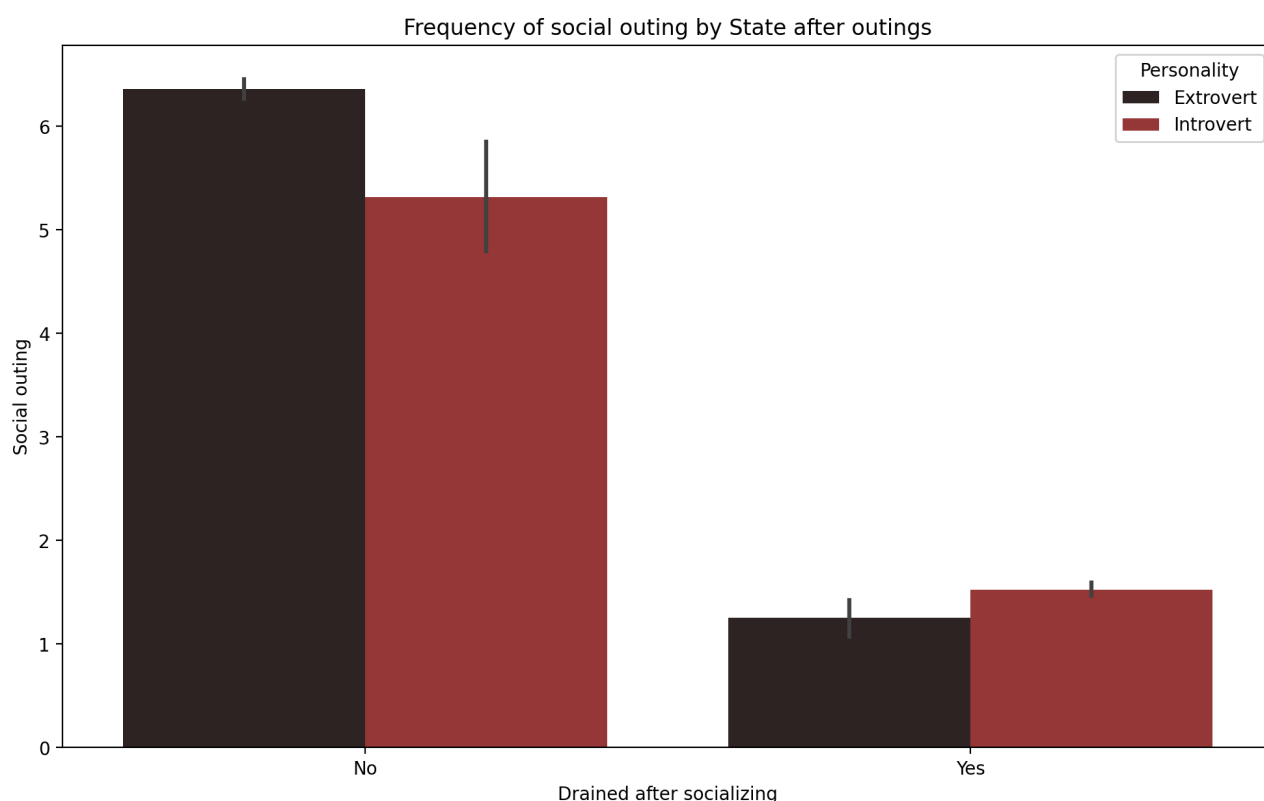
Out [32]:



When looking at this visualization, the boxplot shows the mean, median, outliers, max and minimum values to further understand the data. When looking at the boxplot, it shows that those who feel drained have greater variability in time spent alone as well as many outliers, while individuals who reported as not drained have a uniform distribution.

```
In [33]: sns.barplot(data=hi_new, x='Drained_after_socializing',
y='Social_event_attendance', hue='Personality', palette='dark:brown')
plt.title('Frequency of social outing by State after outings')
plt.xlabel('Drained after socializing')
plt.ylabel('Social outing')
plt.savefig('Project2_Graphs/Frequency_of_social_outing
by_State_after_outings.jpg')
plt.show()
```

Out [33]:



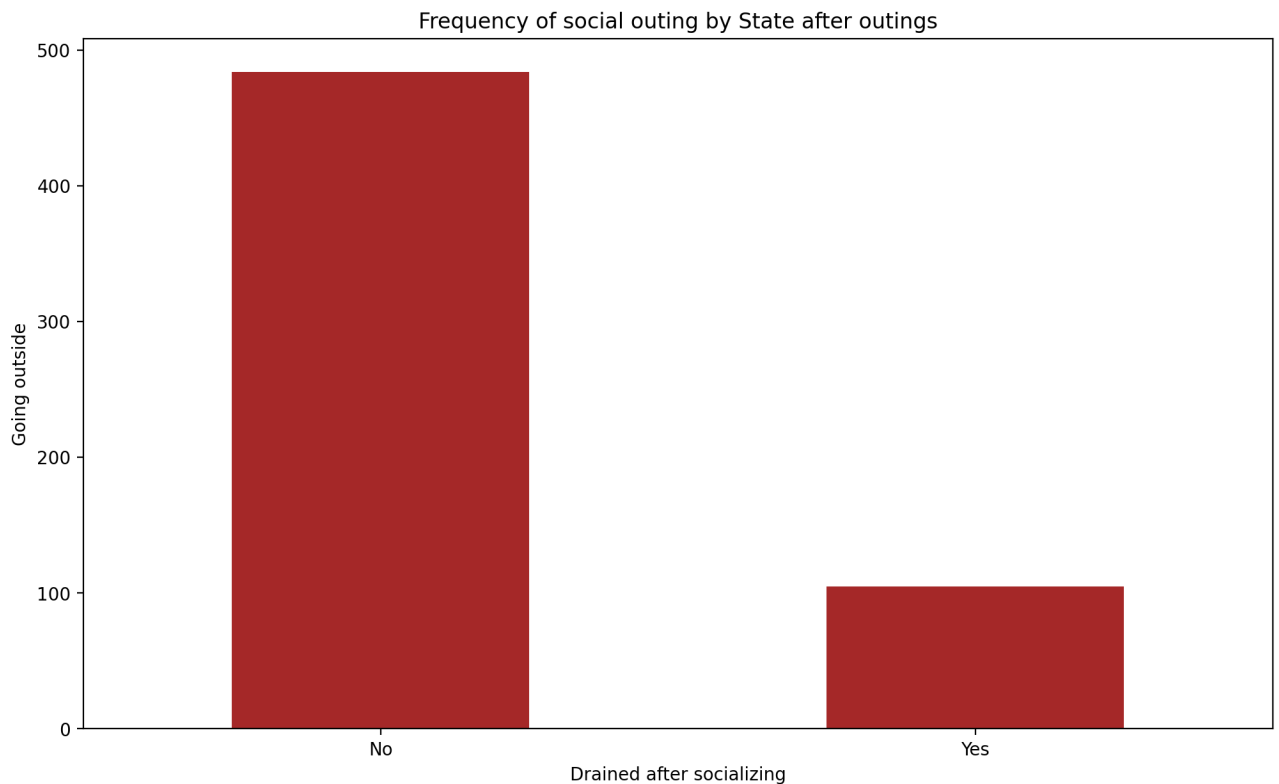
Explain

When looking the bar plot, it shows the average and the different values when comparting both introverts and extroverts. Those who report the same personality within the group have similar ranges while introverts and extroverts have differing results. Both introverts and extroverts have an increase in social outing when they are not drained after socializing; however, when comparing therir results, extroverts are significantly more likely to engage in social outings. This shows that regardless of ones personality type, draining of social energy impacts social outings.

In [34]:

```
state_after_outings = hi_new.groupby('Drained_after_socializing',
observed=True)['Going_outside'].mean() * 100
state_after_outings.plot(kind='bar', color='brown')
plt.title('Frequency of social outing by State after outings')
plt.xlabel('Drained after socializing')
plt.ylabel('Going outside')
plt.xticks(rotation=0)
plt.savefig('Project2_Graphs/group_drained_after_socializing
by_the_by_the_going_outside_avg.jpg')
plt.show()
```

Out [34]:



Explain

This bar chart visualizes the frequency of a persons state after they have a social interaction in public. When looking at the average values, it is evident that there is a division of scores. Individuals who feel drained after socializing have less activity that is spread within the graph and data. Alternately, those who were not drained did not have much spread. It is revealed within this data set that being drained is influential within our data set and predicts that an individual spends more alone time, has less friend groups, and less social interactionnswith the outside world.

Section 2: Relationship Visualizations

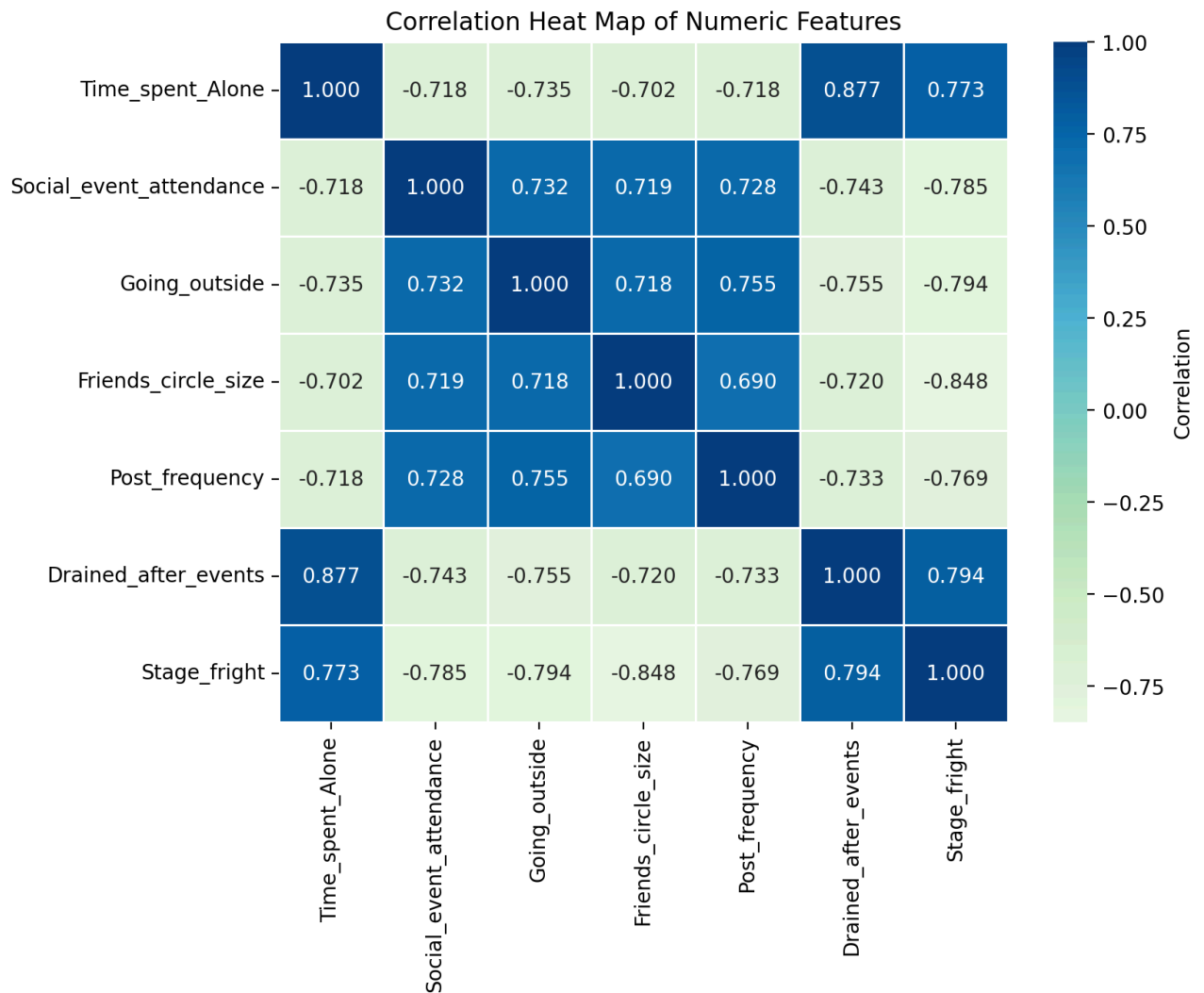
In [35]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [36]:

```
plt.figure(figsize=(8,6))
corr_matrix=hi_new.corr(numeric_only=True)
ax=sns.heatmap(corr_matrix, annot=True, fmt='.3f', cmap='GnBu', center=0,
linewidth=0.5, cbar_kws={'label': 'Correlation'})
plt.title('Correlation Heat Map of Numeric Features')
plt.savefig('Project2_Graphs/Heatmap.jpg')
plt.show()
```

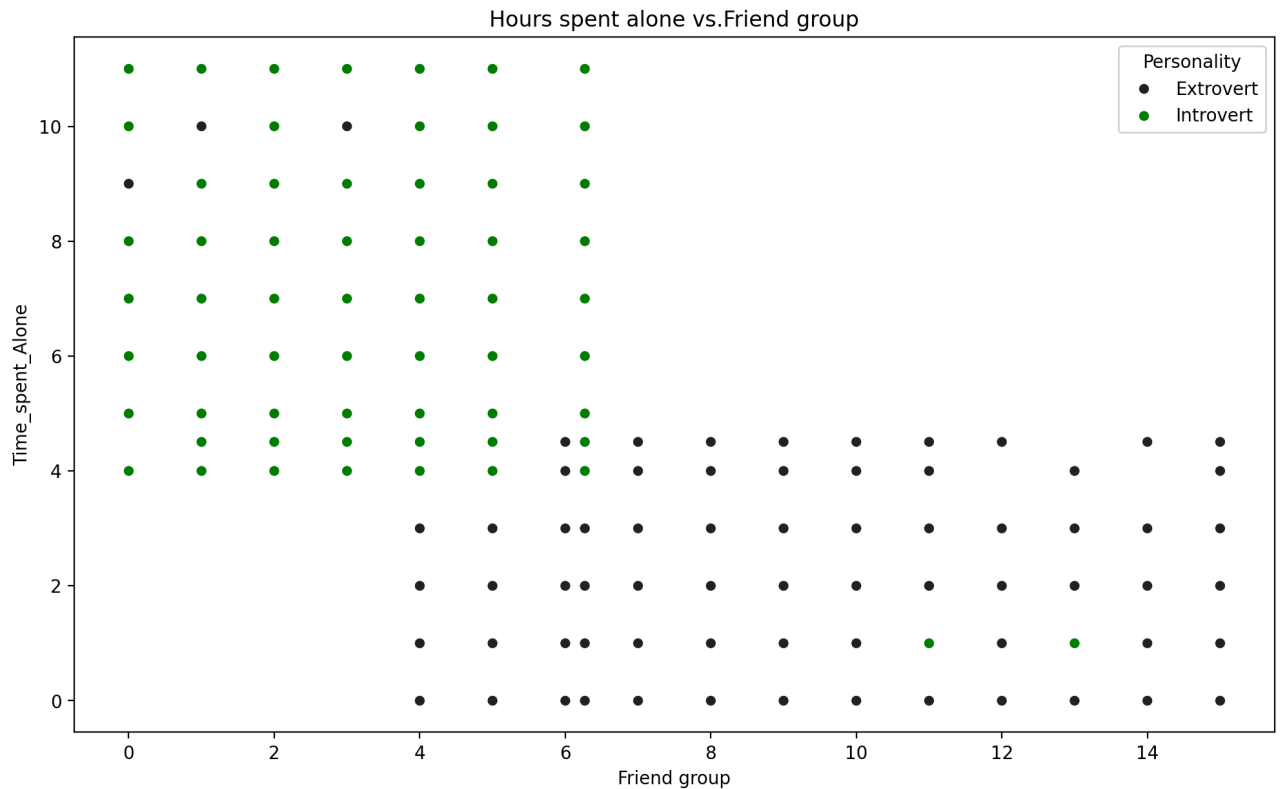
Out [36]:



The heatmap shows uses colors in order to visualize the strengt and relationship between our tested variables. Dark blue colors within our matrix represent a strong positive relationship, meaning when one variable increases so does the other one. The lighter green represents values that are closer to zero where there may not be a strong relationship or one at all. For example a strong negative correlation is seen in our variables time_spent_alne and social_event_attendance (-0.718). This means that the more alone time someone speds, their social event attendance decreases. This is important within our data because it shows how social behaviors are correlated with other factors such as an individuals alone time that they are comfortable with.

```
In [37]: sns.scatterplot(data=hi_new, y= 'Time_spent_Alone',
x= 'Friends_circle_size', hue='Personality', palette= 'dark:green')
plt.title('Hours spent alone vs.Friend group')
plt.xlabel('Friend group')
plt.savefig('Project2_Graphs/Hours_spent_alone_vs._Friend_group_scatterpl
ot.jpg')
plt.show
```

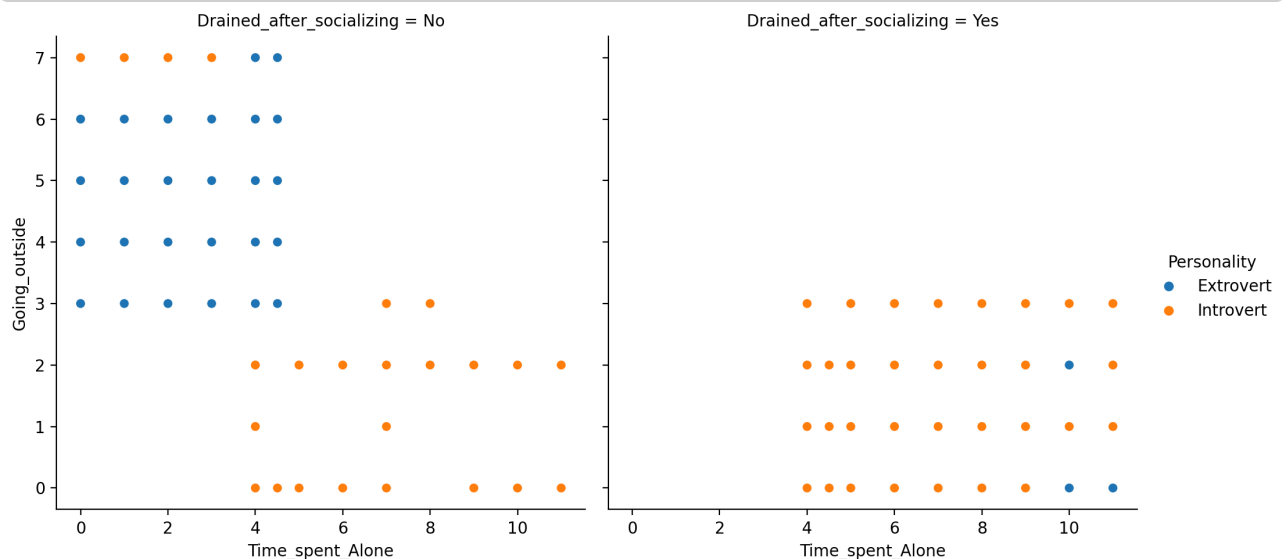
```
Out [37]: <function matplotlib.pyplot.show(close=None, block=None)>
```



When analyzing this scatterplot, it shows the relationship between an individual's alone time and their friend group size. One side of the scatterplot shows individuals engaging in up to 7 hours alone and also having a smaller friend group size, while the other side displaying individuals who spend very small amount of time alone and having a larger friend group size. This visualization supports our research question as it displays a pattern that when an individual's friend group size is smaller, alone time decreases and vice versa.

In [38]: `sns.relplot(data=hi_filtered1, x="Time_spent_Alone", y="Going_outside", hue="Personality", col="Drained_after_socializing", col_wrap=2)
plt.savefig("Project2_Graphs/relplot_time_vs_outside_by_drained.jpg")`

Out [38]:



When loo

Summary and Conclusion

When looking at our dataset, it is evident that social behavior within an individual is influenced by their personality type and the energy they exume. When looking at most of the 2,900 participant values, it is evident that most have smaller friend groups while a smaller amout may have friend groups that have more people. We found it interesting within our data seeing that there was a negative relationship between between nes alone time and the social activities they engage in. When an individual spends more alone time both social engagement/outings and friend group size decrease. Something that we must highlight within our data is the fact that social drainage influences behavior regardless of personalit type. When both introverts and extroverts feel socially drained, they spend more alone time an put themselves into less social engagements.

In Conclusion, our study shows how human interaction is influenced by other determining facrors as well as an individuals comfortableness. Ones social environment as well as innate biological feautres play a large role in human behavior. An individuals personality explains the biological and social processes that in turn, influence their reactions in social outings, interactions, etc. Individuals who are more biologically recharged feel more of a connection towards the social world, while some may not. The visualizations we have provided within our data support the idea that environemtns play a large role for huan interaction; however, our bioloigcal desire also play a large role in our social lives as well.

In [0]: